
PB-2125

整机环境可靠性测试报告

测试工程师：李红勇

审核：王朋刚

核准：刘松辉

测试基本信息

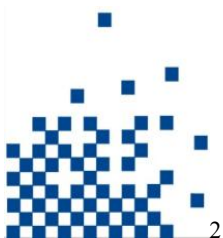


PCB 版本编号	PB-2125
机构编号	2125
芯片方案	RTL8382L+RTL8218D
样机规格	16G+2SFP
软件版本	NETCORE_RTL8382L_16G_2SFP_18D_DISEEE_GPIO1_VLANV_2.23_20230216.BIN
样机数量	5 pcs
测试地点	实验室
测试设备	万用表，示波器，电流碳棒，频谱仪，温箱
硬件工程师	王胜
测试日期	2023-02-20
评审日期	
整机正反面图	 

测试记录

1. 低温贮存

贮存条件与试验模型	测试结果
1、试验参数：温度 $\leq -40^{\circ}\text{C}$ ，贮存时间 ≥ 24 小时 2、样品状态：整机空载下电贮存 3、判定标准：样机外观、结构正常；试验后样品全载配置测试，性能正常，无错误记录	PASS



试验图片

性能测试

' Chassis	' Tx Stream Frame	' Rx Signature S	' Tx Frame	' Rx Frame
Chassis1(BigTao6000):127.0.0.1				
Card1:T6008C 8xGE Copper				
Port4	101,316,741	101,316,664	0	0
Port5	101,316,730	101,316,741	0	0
Port7	101,316,723	101,316,730	0	0
Port8	101,316,719	101,316,723	0	0
Card2:T6008C 8xGE Copper				
Port1	101,316,679	101,316,719	0	0
Port2	101,316,676	101,316,679	0	0
Port3	101,316,670	101,316,676	0	0
Port4	101,316,669	101,316,670	0	0
Port6	101,316,662	101,316,669	0	0
Port7	101,316,662	101,316,662	0	0
Port8	101,316,657	101,316,662	0	0
Card3:T6008C 8xGE Copper				
Port1	101,316,690	101,316,657	0	0
Port2	101,316,687	101,316,690	0	0
Port3	101,316,682	101,316,687	0	0
Port4	101,316,679	101,316,682	0	0
Port5	101,316,673	101,316,679	0	0
Port7	101,316,671	101,316,673	0	0
Port8	101,316,664	101,316,671	0	0
Start At:2023-03-16 19:12:16				
Run Time:000:02:36				



2.高温贮存

贮存条件与试验模型		测试结果																																																																																																																																																																															
1、试验参数：试验温度 $\geq 70^{\circ}\text{C}$ ，贮存时间 ≥ 24 小时 2、样品状态：裸机空载下电贮存 3、指标要求：样机外观、结构正常；试验后样品全载配置测试，性能正常，无错误记录		PASS																																																																																																																																																																															
试验图片																																																																																																																																																																																	
性能测试	<table><tr><th></th><th>' Chassis</th><th>' Tx Stream Fr</th><th>' Rx Signature</th><th>' Tx Frame</th><th>' Rx Frame</th><th>' Rx Sequ</th></tr><tr><td></td><td colspan="6">Chassis1(BigTao6000):127.0.0.1</td></tr><tr><td></td><td colspan="6">Card1:T6008C 8xGE Copper</td></tr><tr><td></td><td>Port4</td><td>53,660,975</td><td>53,660,902</td><td>0</td><td>0</td><td>N/A</td></tr><tr><td></td><td>Port5</td><td>53,660,964</td><td>53,660,975</td><td>0</td><td>0</td><td>N/A</td></tr><tr><td></td><td>Port7</td><td>53,660,957</td><td>53,660,964</td><td>0</td><td>0</td><td>N/A</td></tr><tr><td></td><td>Port8</td><td>53,660,954</td><td>53,660,957</td><td>0</td><td>0</td><td>N/A</td></tr><tr><td></td><td colspan="6">Card2:T6008C 8xGE Copper</td></tr><tr><td></td><td>Port1</td><td>53,660,931</td><td>53,660,954</td><td>0</td><td>0</td><td>N/A</td></tr><tr><td></td><td>Port2</td><td>53,660,928</td><td>53,660,931</td><td>0</td><td>0</td><td>N/A</td></tr><tr><td></td><td>Port3</td><td>53,660,922</td><td>53,660,928</td><td>0</td><td>0</td><td>N/A</td></tr><tr><td></td><td>Port4</td><td>53,660,921</td><td>53,660,922</td><td>0</td><td>0</td><td>N/A</td></tr><tr><td></td><td>Port6</td><td>53,660,915</td><td>53,660,921</td><td>0</td><td>0</td><td>N/A</td></tr><tr><td></td><td>Port7</td><td>53,660,914</td><td>53,660,915</td><td>0</td><td>0</td><td>N/A</td></tr><tr><td></td><td>Port8</td><td>53,660,910</td><td>53,660,914</td><td>0</td><td>0</td><td>N/A</td></tr><tr><td></td><td colspan="6">Card3:T6008C 8xGE Copper</td></tr><tr><td></td><td>Port1</td><td>53,660,926</td><td>53,660,910</td><td>0</td><td>0</td><td>N/A</td></tr><tr><td></td><td>Port2</td><td>53,660,923</td><td>53,660,926</td><td>0</td><td>0</td><td>N/A</td></tr><tr><td></td><td>Port3</td><td>53,660,919</td><td>53,660,923</td><td>0</td><td>0</td><td>N/A</td></tr><tr><td></td><td>Port4</td><td>53,660,916</td><td>53,660,919</td><td>0</td><td>0</td><td>N/A</td></tr><tr><td></td><td>Port5</td><td>53,660,910</td><td>53,660,916</td><td>0</td><td>0</td><td>N/A</td></tr><tr><td></td><td>Port7</td><td>53,660,908</td><td>53,660,910</td><td>0</td><td>0</td><td>N/A</td></tr><tr><td></td><td>Port8</td><td>53,660,902</td><td>53,660,908</td><td>0</td><td>0</td><td>N/A</td></tr><tr><td></td><td colspan="6">Start At:2023-03-16 19:12:16</td></tr><tr><td></td><td colspan="6">Run Time:000:01:24</td></tr></table>			' Chassis	' Tx Stream Fr	' Rx Signature	' Tx Frame	' Rx Frame	' Rx Sequ		Chassis1(BigTao6000):127.0.0.1							Card1:T6008C 8xGE Copper							Port4	53,660,975	53,660,902	0	0	N/A		Port5	53,660,964	53,660,975	0	0	N/A		Port7	53,660,957	53,660,964	0	0	N/A		Port8	53,660,954	53,660,957	0	0	N/A		Card2:T6008C 8xGE Copper							Port1	53,660,931	53,660,954	0	0	N/A		Port2	53,660,928	53,660,931	0	0	N/A		Port3	53,660,922	53,660,928	0	0	N/A		Port4	53,660,921	53,660,922	0	0	N/A		Port6	53,660,915	53,660,921	0	0	N/A		Port7	53,660,914	53,660,915	0	0	N/A		Port8	53,660,910	53,660,914	0	0	N/A		Card3:T6008C 8xGE Copper							Port1	53,660,926	53,660,910	0	0	N/A		Port2	53,660,923	53,660,926	0	0	N/A		Port3	53,660,919	53,660,923	0	0	N/A		Port4	53,660,916	53,660,919	0	0	N/A		Port5	53,660,910	53,660,916	0	0	N/A		Port7	53,660,908	53,660,910	0	0	N/A		Port8	53,660,902	53,660,908	0	0	N/A		Start At:2023-03-16 19:12:16							Run Time:000:01:24					
	' Chassis	' Tx Stream Fr	' Rx Signature	' Tx Frame	' Rx Frame	' Rx Sequ																																																																																																																																																																											
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	Port4	53,660,975	53,660,902	0	0	N/A																																																																																																																																																																											
	Port5	53,660,964	53,660,975	0	0	N/A																																																																																																																																																																											
	Port7	53,660,957	53,660,964	0	0	N/A																																																																																																																																																																											
	Port8	53,660,954	53,660,957	0	0	N/A																																																																																																																																																																											
	Card2:T6008C 8xGE Copper																																																																																																																																																																																
	Port1	53,660,931	53,660,954	0	0	N/A																																																																																																																																																																											
	Port2	53,660,928	53,660,931	0	0	N/A																																																																																																																																																																											
	Port3	53,660,922	53,660,928	0	0	N/A																																																																																																																																																																											
	Port4	53,660,921	53,660,922	0	0	N/A																																																																																																																																																																											
	Port6	53,660,915	53,660,921	0	0	N/A																																																																																																																																																																											
	Port7	53,660,914	53,660,915	0	0	N/A																																																																																																																																																																											
	Port8	53,660,910	53,660,914	0	0	N/A																																																																																																																																																																											
	Card3:T6008C 8xGE Copper																																																																																																																																																																																
	Port1	53,660,926	53,660,910	0	0	N/A																																																																																																																																																																											
	Port2	53,660,923	53,660,926	0	0	N/A																																																																																																																																																																											
	Port3	53,660,919	53,660,923	0	0	N/A																																																																																																																																																																											
	Port4	53,660,916	53,660,919	0	0	N/A																																																																																																																																																																											
	Port5	53,660,910	53,660,916	0	0	N/A																																																																																																																																																																											
	Port7	53,660,908	53,660,910	0	0	N/A																																																																																																																																																																											
	Port8	53,660,902	53,660,908	0	0	N/A																																																																																																																																																																											
	Start At:2023-03-16 19:12:16																																																																																																																																																																																
	Run Time:000:01:24																																																																																																																																																																																



3.低温工作

贮存条件与试验模型		测试结果
1、试验参数：工作温度-10℃，试验工作时间≥24 小时 2、样品状态：满载工作 3、指标要求：样机外观、结构正常；试验过程样品应正常工作，无错误记录（丢包率符合要求、无错包等现象）		PASS
试验图片		

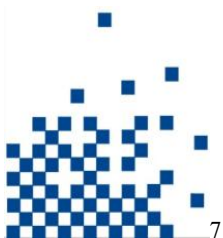


	' Chassis	' Tx Stream Frames	' Rx Signature Str	' Tx Frame	' Rx Frame	' R
性能测试	Chassis1(BigTao6000):127.0.0.1					
	Card1:T6008C 8xGE Copper					
	Port3	72,896,656,140	72,896,655,683	0	0	N/
	Port4	72,896,656,125	72,896,656,140	0	0	N/
	Port5	72,896,656,098	72,896,656,125	0	0	N/
	Port6	72,896,656,074	72,896,656,098	0	0	N/
	Port7	72,896,656,040	72,896,656,074	0	0	N/
	Port8	72,896,656,025	72,896,656,040	0	0	N/
	Card2:T6008C 8xGE Copper					
	Port1	72,896,625,652	72,896,656,025	0	0	N/
	Port2	72,896,625,641	72,896,625,652	0	0	N/
	Port3	72,896,625,632	72,896,625,641	0	0	N/
	Port5	72,896,625,628	72,896,625,632	0	0	N/
	Port7	72,896,625,624	72,896,625,628	0	0	N/
	Port8	72,896,625,619	72,896,625,624	0	0	N/
	Card3:T6008C 8xGE Copper					
	Port1	72,896,655,798	72,896,625,619	0	0	N/
	Port2	72,896,655,790	72,896,655,798	0	0	N/
	Port3	72,896,655,786	72,896,655,790	0	0	N/
	Port4	72,896,655,782	72,896,655,786	0	0	N/
	Port7	72,896,655,699	72,896,655,782	0	0	N/
	Port8	72,896,655,683	72,896,655,699	0	0	N/
	Start At:2023-03-09 09:39:26					
	Run Time:024:00:26					



4. 高温工作

贮存条件与试验模型		测试结果
1、试验参数：工作温度 50℃，试验工作时间≥72 小时 2、样品状态：满载工作 3、判定标准：样机外观、结构正常；试验过程样品应正常工作，无重启，无错误记录（丢包率符合要求、无错包等现象）		PASS
试验图片		

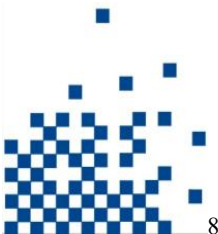


	A	B	C	D		E
1	' Chassis	' Tx Stream Frames	' Rx Signature Str	' Tx Frame	' Rx Fr	
2	Chassis1(BigTao6000):127.0.0.1					
3	Card1:T6008C 8xGE Copper					
4	Port3	222,593,837,702	222,593,839,696	0	0	
5	Port4	222,593,837,698	222,593,837,702	0	0	
6	Port5	222,593,837,694	222,593,837,698	0	0	
7	Port6	222,593,837,687	222,593,837,694	0	0	
8	Port7	222,593,837,672	222,593,837,687	0	0	
9	Port8	222,593,837,662	222,593,837,672	0	0	
10	Card2:T6008C 8xGE Copper					
11	Port1	222,593,747,694	222,593,837,662	0	0	
12	Port2	222,593,747,688	222,593,747,694	0	0	
13	Port3	222,593,747,684	222,593,747,688	0	0	
14	Port5	222,593,747,680	222,593,747,684	0	0	
15	Port7	222,593,747,676	222,593,747,680	0	0	
16	Port8	222,593,747,673	222,593,747,676	0	0	
17	Card3:T6008C 8xGE Copper					
18	Port1	222,593,839,713	222,593,747,673	0	0	
19	Port2	222,593,839,710	222,593,839,713	0	0	
20	Port3	222,593,839,706	222,593,839,710	0	0	
21	Port4	222,593,839,703	222,593,839,706	0	0	
22	Port7	222,593,839,698	222,593,839,703	0	0	
23	Port8	222,593,839,696	222,593,839,698	0	0	
24						
25	Start At:2023-03-10 12:26:03					
26	Run Time:073:13:18					
27						

性能测试

5. 高温高湿工作

贮存条件与试验模型	测试结果
1、环境温度（湿度）：温度 50℃（93%RH）， 试验工作时间≥24 小时 2、样品状态：满载工作 3、判定标准：样机外观、结构正常；试验过程 样品应正常工作，无重启，无错误记录（丢包率符合 要求、无错包等现象）	PASS

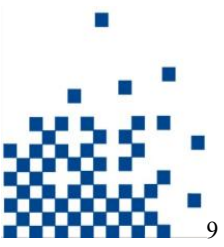


试验图片

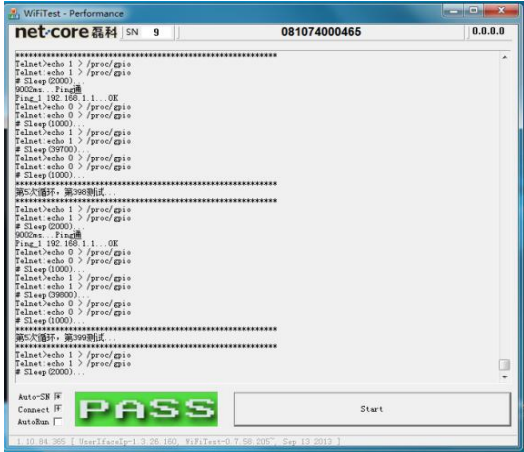


性能测试

' Chassis	' Tx Stream Frames	' Rx Signature Str	' Tx Frame	' Rx Frame	' Rx Sequ
Chassis1(BigTao6000):127.0.0.1					
Card1:T6008C 8xGE Copper					
Port3	76,206,524,276	76,206,235,156	0	0	N/A
Port4	76,206,524,274	76,206,524,276	0	0	N/A
Port5	76,206,524,266	76,206,524,274	0	0	N/A
Port6	76,206,524,261	76,206,524,266	0	0	N/A
Port7	76,206,524,261	76,206,524,261	0	0	N/A
Port8	76,206,524,261	76,206,524,261	0	0	N/A
Card2:T6008C 8xGE Copper					
Port1	76,206,493,275	76,206,524,261	0	0	N/A
Port2	76,206,493,273	76,206,493,275	0	0	N/A
Port3	76,206,493,268	76,206,493,273	0	0	N/A
Port7	76,206,493,254	76,206,493,268	0	0	N/A
Port8	76,206,493,251	76,206,493,254	0	0	N/A
Card3:T6008C 8xGE Copper					
Port1	76,206,425,153	76,206,493,251	0	0	N/A
Port2	76,206,425,148	76,206,425,153	0	0	N/A
Port3	76,206,425,135	76,206,425,148	0	0	N/A
Port4	76,206,425,135	76,206,425,135	0	0	N/A
Port7	76,206,367,628	76,206,425,135	0	0	N/A
Port8	76,206,325,128	76,206,367,628	0	0	N/A
Card4:T6108M 8xGE Dual Media					
Port3	76,206,235,156	76,206,325,128	0	0	N/A
Start At:2023-03-06 16:59:55					
Run Time:025:06:13					



6. 上下电测试

贮存条件与试验模型		测试结果
1、常温或高低温条件下设备上电，在测试程序未加载完成前的随机时间下电，等待 1 秒后重新上电，等测试程序加装完毕，并用打流设备进行发包或 ping 包 10s，设备下电 1s，重新上电，重复以上步骤，循环 1000 次以上（测试时间大等于 24 小时） 2、判定标准：测试设备上下电正常，LOG 正常，收发包相等或正常；试验结束后性能测试正常		PASS
试验图片		
测试附件：  Performance-Pa ss-SN0221-0810		